1.3 NONLINEAR RESISTORS

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Resistance values detailed above are a constant and do not change if the voltage or current-flow alters. But there are circuits that require resistors to change value with a change in temperate or light. This function may not be linear, hence the name**NONLINEAR RESISTORS.**

There are several types of nonlinear resistors, but the most commonly used include : *NTC* resistors (figure a) (Negative Temperature Co-efficient) – their resistance lowers with temperature rise. *PTC* resistors (figure b) (Positive Temperature Co-efficient) – their resistance increases with the temperature rise. *LDR* resistors (figure c) (Light Dependent Resistors) – their resistance lowers with the increase in light. *VDR* resistors (Voltage dependent Resistors) – their resistance critically lowers as the voltage exceeds a certain value. Symbols representing these resistors are shown below.

Fig. 1.4: Nonlinear resistors – a. NTC, b. PTC, c. LDR

In amateur conditions where nonlinear resistor may not be available, it can be replaced with other components. For example, *NTC* resistor may be replaced with a transistor with a trimmer potentiometer, for adjusting the required resistance value. Automobile light may play the role of *PTC*resistor, while *LDR* resistor could be replaced with an open transistor. As an example, figure on the right shows the 2N3055, with its upper part removed, so that light may fall upon the crystal inside.